EDUPRO PROJECT GUIDE

Let’s understand the learning dashboard

A friendly tour of the app, its data, and the analytics ideas inside it

A workbook with 3,000 learners, 60 courses, 60 teachers, and 10,000 enrollment records.

Made to help you explain what the numbers say, how they were calculated, and where their limits are.

1. What did we build?

Think of the dashboard as a set of windows looking at the same school data. The sidebar lets you choose dates, age groups, genders, categories, levels, and free or paid courses. Every chart responds to those choices. That makes it easier to ask a focused question, like: “For this date range, which category had the most enrollments?”

The app is made with Python and Streamlit. Pandas reads and joins the tables, and Plotly draws the interactive charts. The Excel workbook is the source of truth. When its file changes, the app notices its new modified time and reloads the data.

Dashboard area	What you can learn
Overview	Enrollment trend, age mix, category demand, and free vs paid activity.
Learners	Who enrolled and how many enrollments each learner has.
Courses & revenue	Popular subjects, course levels, transaction amounts, and catalog ratings.
Teachers	Recorded enrollments and ratings by instructor and expertise.
Data quality	Checks for duplicate IDs, broken links, missing dates, and invalid amounts.
Explore & export	Inspect filtered rows and download CSV summaries.

Important: this workbook is a saved snapshot, not a live connection to an online school system. The dashboard can reflect a refreshed workbook, but it cannot show new events that were never added to that file.

2. Meet the four tables

A spreadsheet often stores one kind of thing per table. IDs let us connect those tables without relying on names, which can be misspelled or repeated.

Sheet	One row represents	Example key
Users	One learner and their profile	UserID
Courses	One course and its details	CourseID
Teachers	One teacher and their experience/rating	TeacherID
Transactions	One recorded enrollment	TransactionID

Transactions is the activity table: it has a row for each recorded enrollment. A transaction connects to a learner, course, and teacher through their IDs. Joining means lining up those matching IDs so one enrollment row can also tell us the learner’s age, the course category, and the teacher name.

3. Data analytics, in plain language

An analysis usually has a question, a measurement, a grouping, and a careful explanation. Here are the main ideas used in this project.

Idea	Plain meaning	In this dashboard
Count	How many records meet a rule?	Enrollment count is the number of matching transaction rows.
Unique count	How many different people or things?	Learners means distinct UserIDs, not enrollment rows.
Average	Total divided by number of items.	Average enrollments per learner = enrollments ÷ distinct learners.
Share / percent	A part divided by the whole, times 100.	Paid share = paid enrollment rows ÷ all filtered enrollment rows × 100.
Group by	Split records into useful buckets.	Compare enrollments across age bands or categories.
Time trend	Count records by date bucket.	Transactions are grouped by month to show recorded activity over time.
Revenue sum	Add the amount recorded for each row.	Revenue is the sum of Transactions.Amount; units/currency are not specified.
Data quality check	Look for structure that could break a result.	IDs should be unique and each transaction should find its learner/course/teacher.

A percentage always needs a denominator. For example, “30% chose paid courses” means 30% of the filtered enrollment rows are marked Paid. It does not mean 30% of all people in the world or all learners who saw a course.

Four words that keep analysis honest

Describe: say what is visible in the data. Compare: show how groups differ. Explain: give a reason why it happened. Predict: estimate what may happen later. This workbook supports description and comparison. It does not, by itself, prove causes or reliably predict future behavior.

4. How to read the headline numbers

Metric	Formula / meaning	Watch out for
Enrollments	Count of filtered transaction rows.	One learner can have more than one row.
Learners	Distinct UserIDs in matching transactions.	Learners with no matching transaction are not counted here.

Metric	Formula / meaning	Watch out for
Average enrollments / learner	Filtered enrollments ÷ distinct filtered learners.	An average can hide a mix of one-time and frequent learners.
Recorded revenue	Sum of filtered Amount values.	No currency is supplied, so use workbook currency units.
Leading category	Category with the most matching rows.	Popularity is not the same as quality or learning impact.

5. What this workbook says today

This workbook covers January 01, 2025 to December 30, 2025. It contains 3,000 learner profiles, 60 courses, 60 teacher profiles, and 10,000 enrollment records. The recorded Amount values add to 911,323.47 workbook currency units. There are 3,597 enrollments attached to paid courses. The most frequent category is Data Science, with 916 enrollment rows. These figures describe the complete workbook before dashboard filters.

Across the transaction table, there are 3,000 different learners, or about 3.33 transaction rows per learner on average. That average is not a promise that a typical learner takes exactly that many courses; individual learners may be above or below it.

The original workbook's ID fields are unique, its tables have no blank cells, and every transaction points to a learner, course, and teacher. Those are good structural signs. A clean spreadsheet still cannot prove that every real-world enrollment was entered correctly.

6. A practical way to use the dashboard

Step	Try this
1 · Start broad	Leave all filters selected and read the date coverage.
2 · Ask one question	For example: which course category had the most activity?
3 · Narrow the view	Choose a date range or a learner group, then compare the chart.
4 · Check the denominator	See whether the chart counts enrollment rows, learners, revenue, or ratings.
5 · Inspect the records	Use Explore & export to see sample rows behind the summary.
6 · Explain with care	Say what the chart shows, and separate that from your guess about why.

Example explanation: "In the selected period, Programming had the most recorded enrollments. This tells us it was the most selected category in this workbook view. It does not tell us why learners selected it." That is a strong data explanation because it is specific, understandable, and does not claim more than the evidence.

7. Things to remember

- Enrollments are transaction rows; they are not the same as unique learners.
- Revenue is recorded transaction amount, not necessarily profit or cash received after refunds.
- Ratings are recorded ratings, not proof of course quality or teacher effectiveness.
- Activity over time can show when enrollments were recorded, but not why they rose or fell.
- A group with more enrollments may simply have more learners or more available courses.
- Filters change the denominator, so always mention the view you are describing.

A final mental model

Data is a map, not the territory. The workbook gives us a structured view of recorded learning activity. Good analysis reads that map carefully, checks how it was drawn, and explains both what it shows and what it cannot show.

Keep learning: try changing one filter at a time and predict which number should move before you look. Then explain why it moved. That small habit builds strong analytical thinking.